MediTrack · System Architecture & UI/UX User Guide

Complete Operational Specification & Technical Reference

1. Product Overview & Core Design Goals

MediTrack is a high-reliability medication reminder and adherence tracking web application designed with health ergonomics, multi-person family support, and responsive touch-first design. The system operates on zero external database dependency by default, leveraging browser-isolated LocalStorage with instant JSON backup export/import and Web Audio synthesizer chimes for alert generation.

2. Key UI/UX Modules & Capabilities

Module	Functional Description & Ergonomics
Today's Schedule	Real-time timeline displaying all scheduled doses grouped chronologically. Features quick status indicators (Due Now, Upcoming, Taken, Snoozed, Skipped, Missed).
7-Day Adherence Rings	Visual radial rings displaying day-by-day compliance rates (K/T). Clicking any ring opens that day's complete medication checklist.
Web Audio Chimes	Gentle harmonic multi-tone synthesizer (C5, E5, G5 arpeggio) built on the Web Audio API with exponential gain release to ensure alerts are audible yet pleasant.
Dose State Machine	Dynamic status resolver accounting for scheduled times, 2-hour grace period for missed doses, custom snooze timers, and instant undo capabilities.
Medicine Cabinet	Comprehensive prescription management with pill form factors (tablets, capsules, liquids, drops, inhalers), food instructions, and stock countdown with 1-click refill.
Simulation Console	Interactive UI test toolbar allowing reviewers and clinicians to fast-forward hours, jump to morning/afternoon/night, and test alert banners immediately.
Data Portability & PDF	Full JSON database export/import and clinical-grade vector PDF export for doctor consultations and pharmacy refills.

3. Dose Lifecycle State Diagram

State	Behavioral Trigger & State Actions
UPCOMING	Current time is earlier than the scheduled dose timestamp. Shows scheduled time in HH:MM.
DUE NOW	Current time reaches scheduled timestamp. Audio chime triggers, banner rings, vibration/push alerted.
SNOOZED	User requested 10-minute postponement. Mutes alert and sets countdown timer.
TAKEN	User confirmed dose ingestion. Stock decrements by 1; adherence score increases; success tone plays.
MISSED	Current time exceeds scheduled time by more than 2 hours without confirmation.
SKIPPED	User intentionally deferred or declined dose (e.g. physician advised fasting). Recorded separately from missed.